

Inference with logistic regression models

Logistics

- + Final part of project due last day of class
 - + No additional analysis needed; just combining work from first two parts and adding introduction, conclusion, etc.
 - + Work time in class on Wednesday and Friday
- + Homework 7 (optional) due Wednesday
- + Final exam: December 8, 9am - 12pm in Carswell 101 (our usual room)
 - + Final exam will be cumulative, and will include logistic regression

Recap: maximum likelihood estimation for logistic regression

Likelihood:

- + For estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, $\hat{\pi} = \frac{\exp\{\hat{\beta}_0 + \hat{\beta}_1 x\}}{1 + \exp\{\hat{\beta}_0 + \hat{\beta}_1 x\}}$
- + $L(\hat{\beta}_0, \hat{\beta}_1) = P(\text{data})$

Maximize:

- + Choose $\hat{\beta}_0, \hat{\beta}_1$ to maximize $L(\hat{\beta}_0, \hat{\beta}_1)$

Example: GPA and med school admission

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

We want to know if there is actually a relationship between GPA and med school admission.

How can I express this as null and alternative hypotheses about one or more model parameters?

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If this was a linear regression model, how could I test these hypotheses?

Example: GPA and med school admission

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$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

If this was a linear regression model, how could I test these hypotheses?

With a nested F test, or with a t -test.

But, this is logistic regression!

Reminder: nested F test

Recall that the nested F test for comparing full and reduced models uses the test statistic

$$F = \frac{\frac{1}{\text{\#parameters tested}} (SSE_{reduced} - SSE_{full})}{\frac{1}{n-p} SSE_{full}}$$

Motivation:

- + We fit a model to minimize SSE
- + We want to compare the fit of the full and reduced models

Can I use this test statistic for logistic regression models?

Reminder: nested F test

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Motivation:

- + We fit a model to minimize SSE
- + We want to compare the fit of the full and reduced models

Can I use this test statistic for logistic regression models?

No -- logistic regression doesn't involve minimizing SSE. It involves maximizing likelihood!

Deviance

Deviance: If L is the likelihood, then deviance is given by $-2 \log L$

- + Maximizing likelihood is equivalent to minimizing deviance
- + Deviance is analogous to SSE from linear regression

We can use deviance to test hypotheses about model parameters.

Example: GPA and med school admission

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

```
med_glm <- glm(Acceptance ~ GPA, data = MedGPA,  
               family=binomial)
```

```
summary(med_glm)
```

deviance of intercept-only model β_0

Null deviance: 75.791 on 54 degrees of freedom
Residual deviance: 56.839 on 53 degrees of freedom

deviance of full model
 $\beta_0 + \beta_1 \text{GPA}$

Comparing deviances

Null deviance: 75.791 on 54 degrees of freedom
Residual deviance: 56.839 on 53 degrees of freedom

75.791 = deviance for intercept-only model $\log\left(\frac{\pi}{1-\pi}\right) = \beta_0$

56.839 = deviance for full model

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

drop-in-deviance: deviance for reduced model - deviance for full model = 18.95

similar to $SSE_{\text{reduced}} - SSE_{\text{full}}$

Comparing deviances

75.791 = deviance for intercept-only model $\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0$

56.839 = deviance for full model

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

drop-in-deviance: deviance for reduced model - deviance for full model = 18.95

Intuition: a larger drop in deviance is stronger evidence for a relationship between GPA and med school acceptance

Comparing deviances

drop-in-deviance: G = deviance for reduced model - deviance for full model = 18.95

Full model: $\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$

Reduced model: $\log\left(\frac{\pi}{1-\pi}\right) = \beta_0$

Why is G always ≥ 0 ?

linear regression:

add variables to model: $SSE \downarrow$
 $\Rightarrow SSE_{\text{reduced}} \geq SSE_{\text{full}}$

logistic:

add variables:
deviance $\downarrow$
 $\Rightarrow deviance_{\text{reduced}} \geq deviance_{\text{full}}$

Comparing deviances

drop-in-deviance: G = deviance for reduced model - deviance for full model = 18.95

Full model: $\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{GPA}$

Reduced model: $\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0$

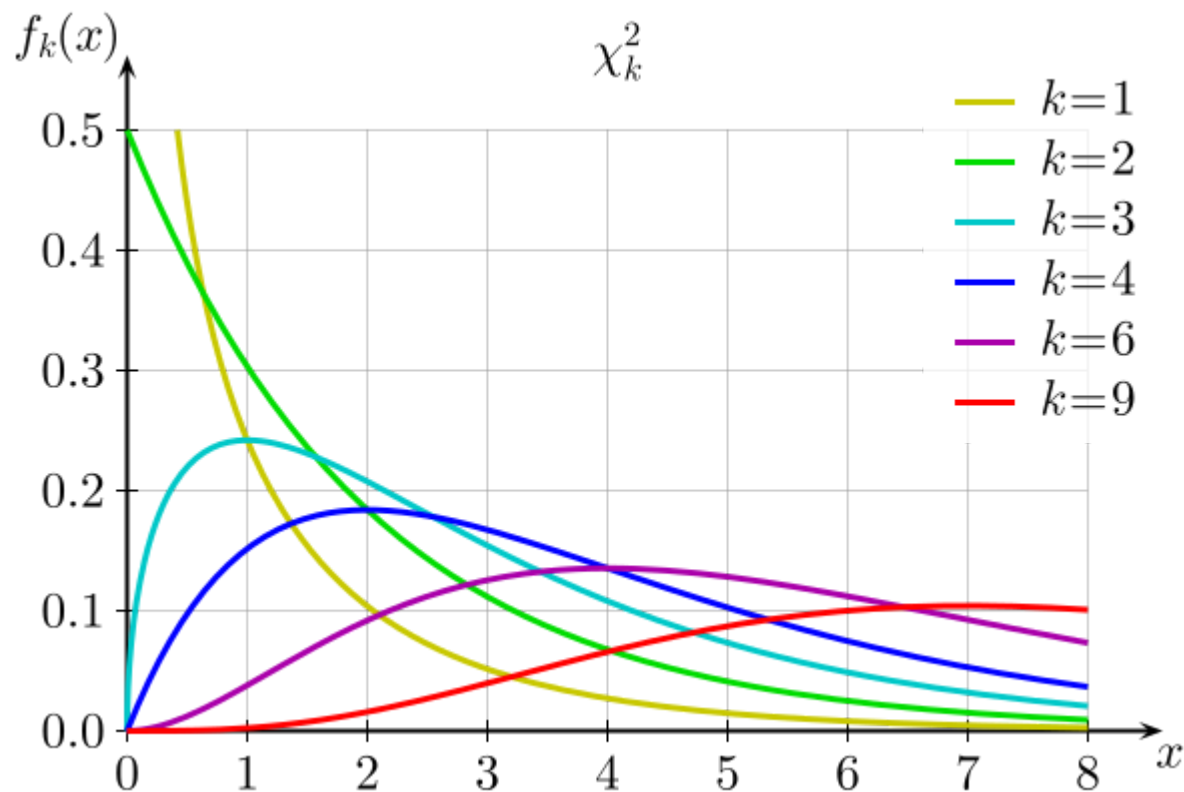
$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

How do I calculate a p-value?

χ^2 distribution

Under H_0 , $G \sim \chi^2_{\# \text{ parameters tested}}$

χ^2_k distribution: parameterized by degrees of freedom k



Computing a p-value

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

G = deviance for reduced model - deviance for full model = 18.95

$$G \sim \chi^2_{\# \text{ parameters tested}}$$

What are my degrees of freedom for the χ^2 distribution in this hypothesis test?

$$\text{df} = 1$$

(testing β_1)

Computing a p-value

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

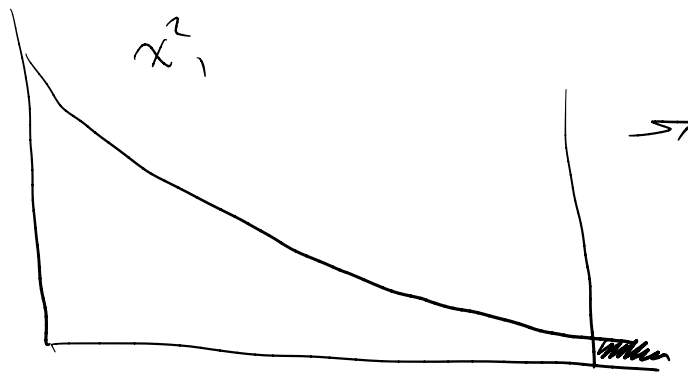
$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

G = deviance for reduced model - deviance for full model = 18.95
 $\sim \chi_1^2$

```
pchisq(18.95, df = 1, lower.tail=FALSE)
```

```
## [1] 1.341893e-05
```

1.341893e-05
p-value



Likelihood ratio test for nested models

+ Compare full and reduced models. Example:

+ full model: $\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{GPA}$

+ reduced model: $\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0$

Likelihood ratio test for nested models

- + Compare full and reduced models
- + Calculate deviance ($-2 \log L$) for full and reduced models.

Example: *reduced*

- + deviance for ~~full~~ model = 75.791

- + deviance for ~~reduced~~ *full* model = 56.839

Likelihood ratio test for nested models

- + Compare full and reduced models
- + Calculate deviance ($-2 \log L$) for full and reduced models
- + Test statistic: G = deviance for reduced model - deviance for full model
 - + Example: $G = 75.791 - 56.839 = 18.95$

Likelihood ratio test for nested models

- + Compare full and reduced models
- + Calculate deviance ($-2 \log L$) for full and reduced models
- + Test statistic: G = deviance for reduced model - deviance for full model
 - + Example: $G = 75.791 - 56.839 = 18.95$
- + p-value: $G \sim \chi^2_{\# \text{ parameters tested}}$
 - + Example: $G \sim \chi^2_1$

Likelihood ratio test: strength and weakness

Strength: Can test multiple parameters at once. E.g.,

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \neq 0$$

Weaknesses:

+ Cannot test one-sided alternative. E.g., can't test

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 > 0$$

+ Cannot test for values other than 0. E.g., can't test

$$H_0 : \beta_1 = 1 \quad H_A : \beta_1 \neq 1$$

z-test

Alternative: Wald tests for single parameters

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

$$H_0 : \beta_1 = b \quad H_A : \beta_1 \neq b$$

+ or $H_A : \beta_1 > b$, or $H_A : \beta_1 < b$

+ b is any value of interest (e.g., 0)

$$z = \frac{\hat{\beta}_1 - b}{SE_{\hat{\beta}_1}} \sim N(0, 1) \text{ (under } H_0 \text{)}$$

*↑ standard normal distribution
(mean = 0, SD = 1)*

Example

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

```
summary(med_glm)
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-19.207	5.629	-3.412	0.000644	***
GPA	5.454	1.579	3.454	0.000553	***

$$z = \frac{5.454}{1.579} = 3.454 \sim N(0, 1)$$

p-value = 0.000553

Wald tests vs. likelihood ratio tests

Wald test

- + like t-tests
- + test a single parameter
- + some example hypotheses:
 - + $H_0 : \beta_1 = 0$ vs.
 $H_A : \beta_1 \neq 0$
 - + $H_0 : \beta_1 = 1$ vs.
 $H_A : \beta_1 > 1$

Likelihood ratio test

- + like nested F-tests
- + test one or more parameters
- + some example hypotheses:
 - + $H_0 : \beta_1 = 0$ vs.
 $H_A : \beta_1 \neq 0$

p-values are different, because test statistics and distributions are different

Class activity

Work on the class activity on the course website. Submit your rendered HTML on Canvas at the end of class.

Class activity

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{ speed}$$

Is there any relationship between whether a pitch is successful and the speed at which it crosses home plate?

What are my null and alternative hypotheses?

Class activity

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$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{ speed}$$

Are faster pitches are more likely to be successful?

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Which test should I use?

Class activity

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Are faster pitches are more likely to be successful?

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 > 0$$

Which test should I use?

Use a Wald test, not a likelihood ratio test, because alternative is one-sided.

Class activity

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 > 0$$

```
kershaw_glm <- glm(Result ~ EndSpeed, data = Kershaw,
                    family=binomial)
summary(kershaw_glm)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.432279	0.488616	-2.931	0.003375	**
EndSpeed	0.023360	0.006019	3.881	0.000104	***

Class activity

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(Intercept)	-1.432279	0.488616	-2.931	0.003375	**
EndSpeed	0.023360	0.006019	3.881	0.000104	***

$$z = \frac{0.0234}{0.006} = 3.881$$

```
pnorm(3.881, lower.tail=F)
```

```
## [1] 5.201392e-05
```